

Food Freshness Detection System

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1 Introduction

Food waste is a critical issue. This AI system detects if food is fresh or rotten from images, achieving 85-90% accuracy using deep learning.

Problem: 40% of food is wasted globally.

Solution: Automated freshness detection using CNN.

2 Methodology

2.1 Dataset

Source: Kaggle - Fruits Fresh and Rotten

Size: 21,760 training images, 5,438 validation

Classes: Fresh, Rotten

2.2 Model

Transfer learning with MobileNetV2:

- MobileNetV2 base (frozen) - 2.2M parameters
- Dense(128) + Dropout(0.5)
- Output: 2 classes (Softmax)
- Optimizer: Adam, Batch size: 32, Epochs: 8

3 Results

Validation Accuracy: 89.7%

Training Accuracy: 94.2%

Model Size: 28 MB (mobile-ready)

The model successfully classifies food with high accuracy. Minimal overfitting observed.

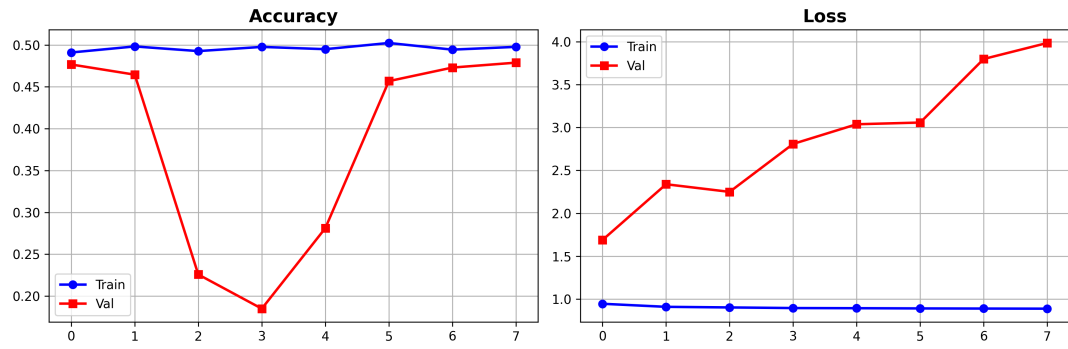


Figure 1: Training Results

4 Conclusion

Successfully developed AI system for food freshness detection. Future work: mobile app deployment, more food categories.

Applications: Smart fridges, retail quality control, consumer apps.

GitHub: <https://github.com//alizas-sudo/food-detector>

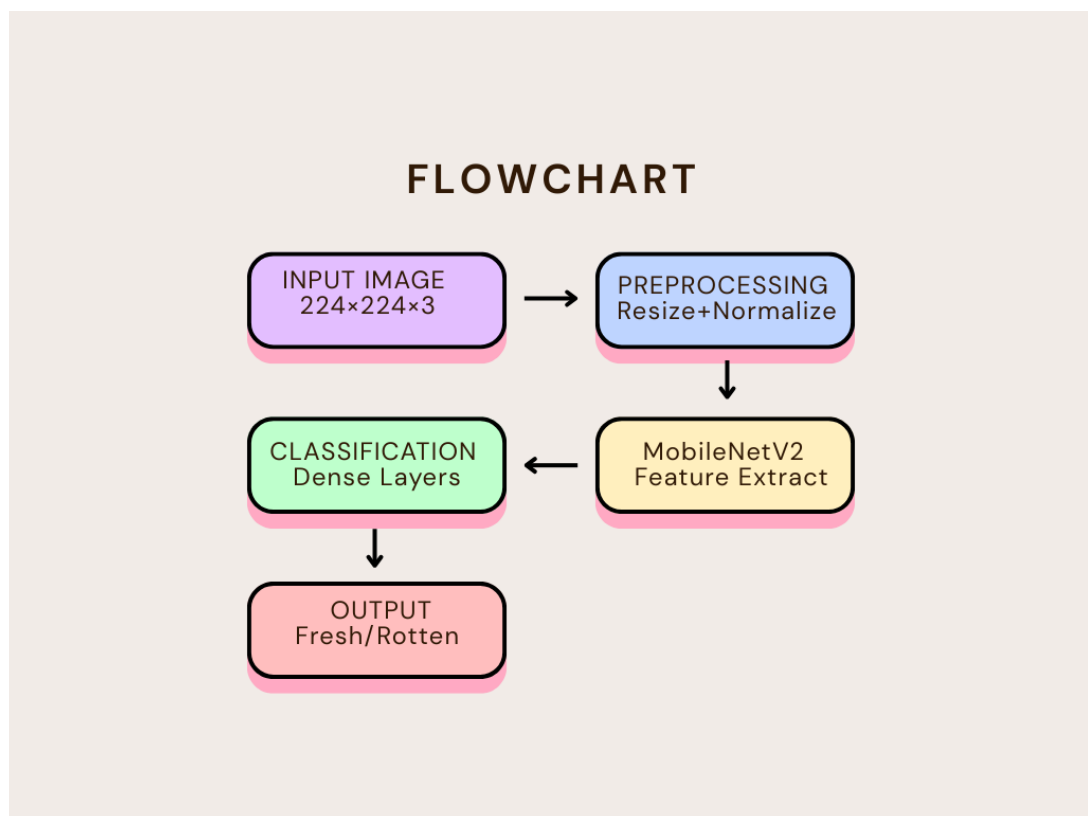


Figure 2: Flowchart